

Investigating the Influence of Sound Source Visualization on the Ventriloquism Effect in an Auralized Virtual Reality Environment

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ABSTRACT

The ventriloquism effect (VQE) describes the illusion that the puppeteer's voice seems to come out of the puppet's mouth. This effect can even be observed in virtual reality (VR) when a spatial discrepancy between the auditory and visual component occurs. However, previous studies have never fully investigated the impact of visual quality on VQE. Therefore, we conducted an exploratory experiment to investigate the influence of the visual appearance of a loudspeaker on the VQE in VR. Our evaluation yields significant differences in the vertical plane which leads to the assumption that the less realistic model had a stronger VQE than the realistic one.

1 INTRODUCTION

Virtual reality (VR) applications let the users immerse into 3D environments and allow them to move freely in a natural way in six degrees of freedom (6-DoF). Compared to non-VR applications, 3D sound (spatial audio) with headphones is an important aspect to enhance the immersion. It allows accurate localization of sound sources and even outer-head localization so-called externalization. To achieve 3D audio the signal needs to be reconstructed with the correct filtering effect of the ear, taking into account the orientation of the listener's head in relation to the sound source. This filtering effect is described by a *head-related transfer function* (HRTF) and is used in modern days VR headsets that are capable of precise 6-DoF tracking. Thereby, an interesting effect can be observed, the *ventriloquism effect* (VQE). It describes the illusion that the voice of the puppeteer seems to come from the puppet's mouth. It is induced when a spatial discrepancy between the auditory and visual component occurs. The origin of the sound is biased towards the visual component. Research revealed that the separation can reach up to 30° on the horizontal plane and can be 2.5 times greater on the vertical plane [4, 5]. Investigating the VQE, 3D audio must be represented via dynamic binaural synthesis with additional auralization to achieve accurate spatial auditory perception [6]. The VQE is well studied and primarily carried out in real environments [3]. However, this effect is not bound to real-life situations though and can be induced in VR too [8]. To create convincing VQE, several factors have to be addressed, like temporal and spatial discrepancy of the audiovisual event, the attention of the user, or the *compellingness* of the audiovisual stimuli [4]. The visualization part in evaluations of VQE is often not mentioned. For example, only short light flashes and noise bursts are used in real environments for visual and auditory stimuli [1, 12]. Few studies address the influence of the visual quality of a sound source on the ventriloquism effect, for example, Warren et al. conclude that high compelling visual stimuli have a more or less strong effect than less compelling ones [11].

Therefore, we conducted an exploratory experiment in VR to evaluate the influence of different visual graduations of a virtual sound source on the ventriloquism effect in a virtual environment.

2 EXPERIMENT

To evaluate the VQE for different degrees of visualization, we conducted an experiment, where each subject shifts the virtual sound source away from the visual representation until the VQE breaks. In our study, the participants wore an HMD (Oculus Rift S) and a non-individualized high-resolution HRTF with 2702 spherical points was used [2]. Unity 3D was used to realize a photorealistic environment while the spatial audio plugin *Steam Audio* recreated the auditive counterpart by a natural precalculated geometry-based reverb (see Fig.1). The *Beyerdynamic DT800 600 Ohm* with an amplifier was used for audio playback. Besides recording how far the sound source was shifted, we used the *Igroup* presence questionnaire [7] extended by custom auditive questions to increase evaluation quality.

2.1 Conditions

Three variations of the visual quality of the loudspeaker were used: Basic - A box like representation (B), Abstract - An untextured speaker (A) and Real - photogrammetry scan of a real loudspeaker (R) (see Fig. 2). The movement was split in two different shift modes: Horizontal (H) and Vertical (V) movement. To increase variation, two anechoic sounds were used: Guitar (G) and Speech (S). The experiment had a within-subject design and each combination was performed four times which results in 48 runs for every participant.

2.2 Procedure & Task

The experiment started with a short training introduction in the virtual environment to make sure that the participants felt confident in VR. Afterwards, each run started with a randomized position and condition of the sound source. The selected position was spawned in front of the subject with a distance of two meters. Based on the shift mode, the random range varied for horizontal and vertical shifts ($\alpha_{AZ} = \pm 25^\circ$, $\alpha_{EL} = \pm 15^\circ$). The subject's task was to move a matching virtual sound source away from its static visual counterpart in a horizontal or vertical direction. For this, they used the joystick of the *Oculus* controller. A reversed task, where the subject should be turning the shifted sound source towards the visual object, was discarded because of the individual specific threshold. The movement was performed in an arc so that the distance always remained

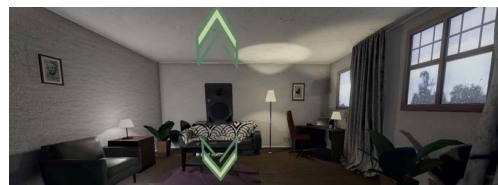


Figure 1: The room as seen by the subjects with the photogrammetry model (R) and the vertical shift mode (V).

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Figure 2: The visual representations of the loudspeaker (left to right): **Basic**, **Abstract**, **Realistic**

the same. This was done to prevent distance-based artifacts. The subjects were instructed to stop when the sound was no longer perceived as coming from the visual loudspeaker. After confirmation, the loudspeaker and its sound source despawned and the subject started the next randomized run.

3 RESULTS

The study involved $n = 11$ people (4 female and 7 male) with an average age of $M = 31.55$ years. Before the results were analyzed, extreme outliers were removed from the data and Shapiro-Wilk significance tests were performed to check for normality. All tests are corrected according to Bonferroni (see Fig. 3).

Filtered by the visualization and shift mode, two *Friedmans* two-factor variance analyses by rank with related samples reveal no significant differences, $\chi^2_H(2) = 2.69$, $p_H = .261$ ($M_{BH} = 30.46^\circ$, $M_{AH} = 27.9^\circ$, $M_{RH} = 28.01^\circ$), $\chi^2_V(2) = 4.3$, $p_V = .116$ ($M_{BV} = 66.7^\circ$, $M_{AV} = 57.6^\circ$, $M_{RV} = 54.9^\circ$). Nevertheless, a pairwise comparison in the vertical mode shows a significant difference between **B** and **R** (Wilcoxon signed ranked test, $Z = -2.57$, $p = .03$). Furthermore, the isolated data by the shift mode ($M_H = 27.4^\circ$, $M_V = 59.9^\circ$) indicates a significant difference as well (Wilcoxon signed ranked test, $Z = 6.047$, $p < .001$). On the other hand, split by the sound playback ($M_G = 36.31^\circ$, $M_S = 35.82^\circ$), a Wilcoxon signed ranked test reveals no significant difference ($Z = -.612$, $p = .541$). The results of the IPQ are $M_{SP} = 4.0$ ($SD_{SP} = 0.93$), $M_{INV} = 2.7$ ($SD_{INV} = 1.5$), $M_{REAL} = 1.98$ ($SD_{REAL} = 1.05$) and $M_G = 3.0$ ($SD_G = 1.48$).

3.1 Discussion

The results indicate no significant differences in visualization quality in horizontal shift mode and as well as in sound overall (**G**, **S**). Interestingly, the vertical shift mode is more affected which is also seen in the results. The visualization quality clearly influences the VQE on the vertical plane. But the shift mode itself also indicates a significant difference on the VQE. We speculate that this relies on monaural cues like the frequency spectrum of the sound which results in a less accurate localization on the vertical plane. Therefore, we conclude that a less recognizable visual loudspeaker representation increases VQE at least in the vertical plane. A realistic model

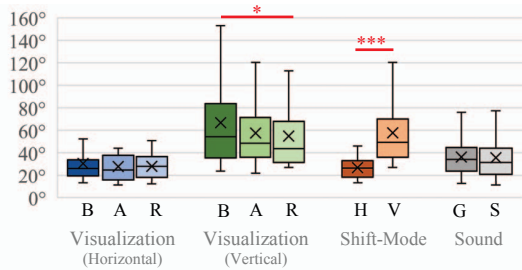


Figure 3: The results of the experiment of the measured angle threshold until the ventriloquism effects breaks. Angles were split by their groups (* : $p < 0.05$, *** : $p < 0.001$).

was more accepted by the subjects as a sound source, while a less appealing increase diffuseness. A more realistic appearance induces a more natural hearing behaviour, that results in less strong VQE.

4 LIMITATIONS & CONCLUSION

A drawback of our system is the evaluation of the audio rendering. It is not clear how the system performs compared to other state-of-the-art sound rendering software. However, the results of the IPQ and sound questionnaire indicate that Steam Audio is suitable and creates a convincing and plausible virtual acoustic environment.

Even though the VQE shows no significant differences in the horizontal plane, our experiment indicates a significant influence of the quality of the visual representation of the sound source in the vertical plane. Despite the expectation, the less compelling model caused a stronger VQE than the realistic one. However, with increasing realism, the subjects were able to distinguish better whether the sound was still coming from the loudspeaker.

For future work, we plan to perform an extended user study with a larger sample size. Follow-up studies could replace the loudspeaker model with visual gradations of human-like avatars [10] or extend the visual variance on the whole virtual environment [9]. It can be concluded that a less compelling visual representation results in a stronger effect and a more blurred localization. Contrary to the assumption that the effect would be stronger with a realistic presentation, we come to a different conclusion which should be further evaluated in future experiments.

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